

Python Programming Final Project

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Predicting Implied
Volatility and Options
Pricing Analysis using
Machine Learning tools



Options and Implied Volatility

- Options are contracts for rights to buy/sell an asset
- Implied volatility - how much the market is expected to move
- Why predict IV?



Dataset overview

- 3 million records of SPY2020-2022
- Model focuses on Call options
- Key columns: strike, underlying price, DTE, Greeks
- Target: c_iv (call option implied volatility)

Black Scholes Pricing Model

Formula

$$C = N(d_1)S_t - N(d_2)Ke^{-rt}$$

$$\text{where } d_1 = \frac{\ln \frac{S_t}{K} + (r + \frac{\sigma^2}{2})t}{\sigma\sqrt{t}}$$

$$\text{and } d_2 = d_1 - \sigma\sqrt{t}$$

C = call option price

N = CDF of the normal distribution

S_t = spot price of an asset

K = strike price

r = risk-free interest rate

t = time to maturity

σ = volatility of the asset

Data Preparation

- Handling missing values and filtering only call options
- Dropping leakage-prone features (e.g., p_iv, c_last)
- Add derived features – moneyness, log_moneyness, tte
- Train/test split

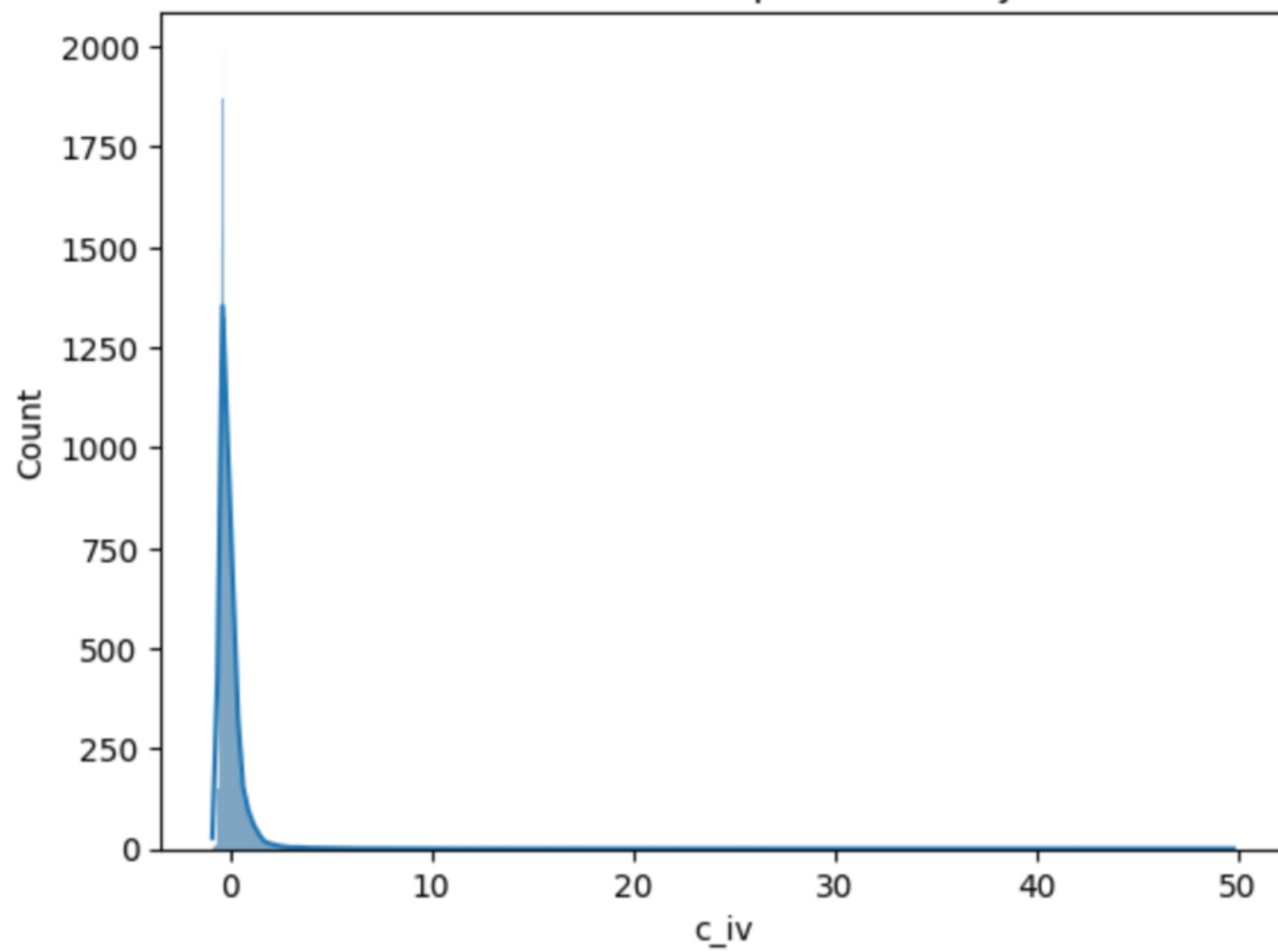
Preprocessing

- Remove outliers using IQR
- Pearson Correlation
- Random Forest Regression choice
- Data Visualization

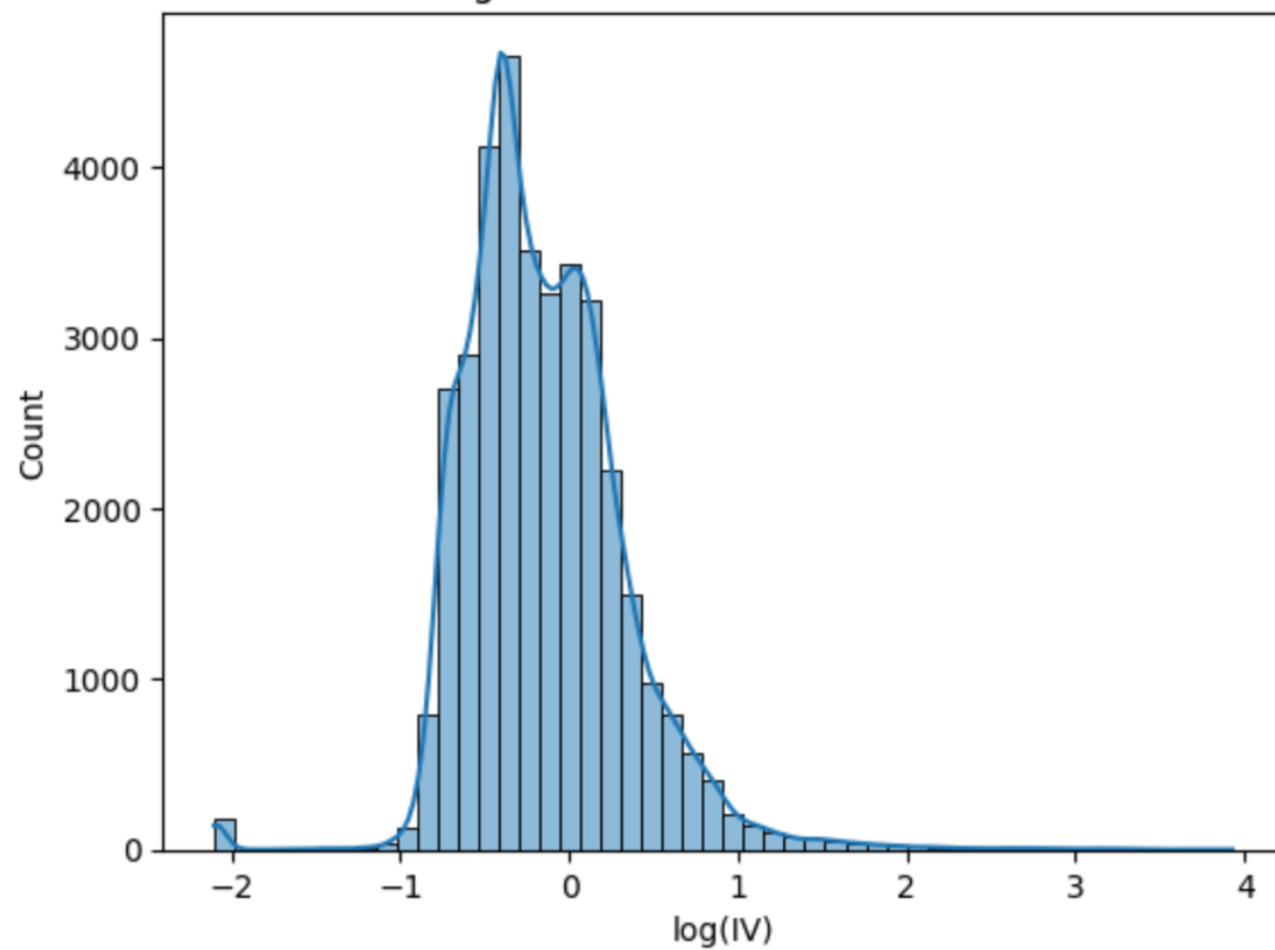
Normalization

- Log transform on target for skewness
- Random forest does not need normalization
- Is necessary for proper visualization
- Standard scaler normalization

Distribution of Implied Volatility

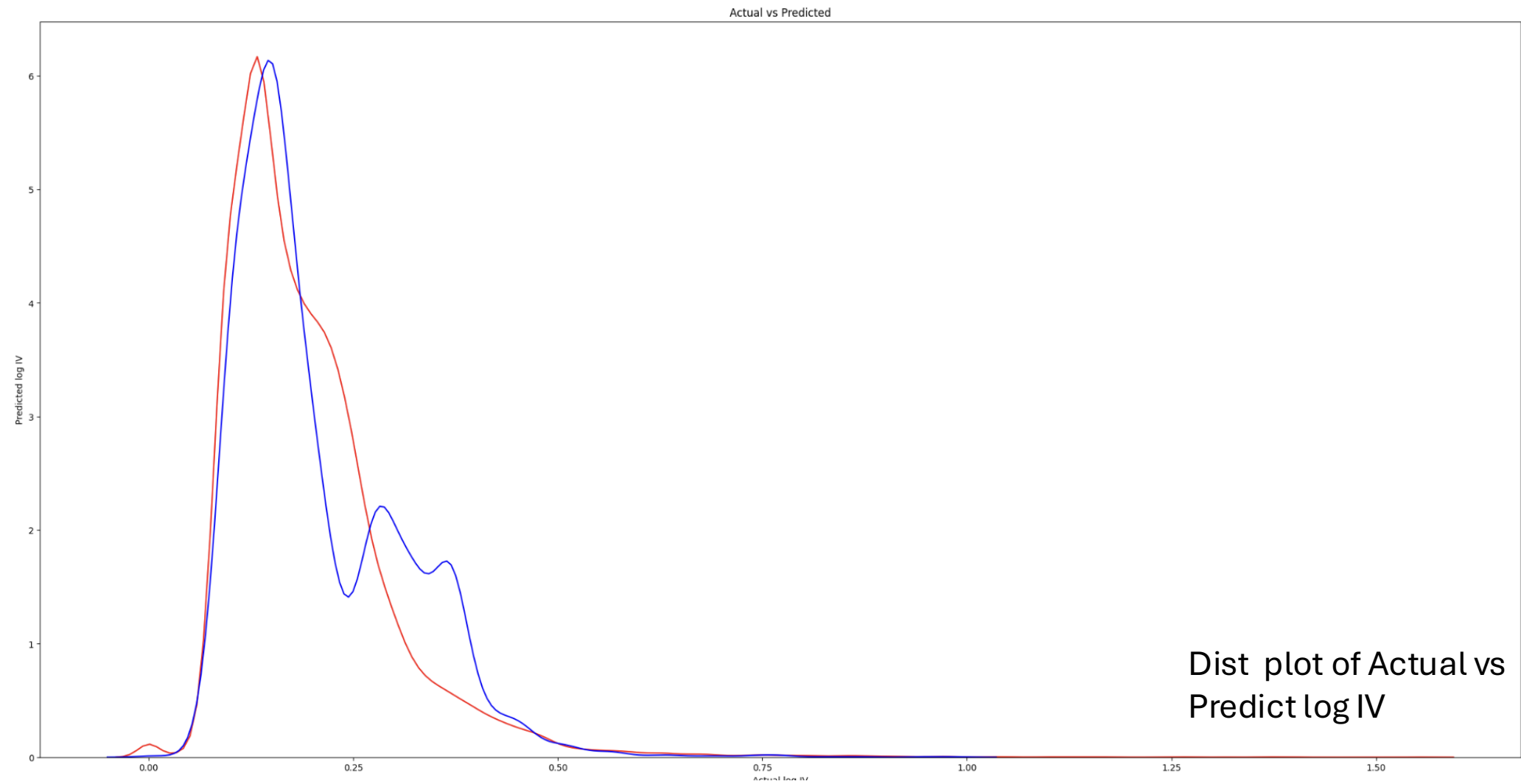


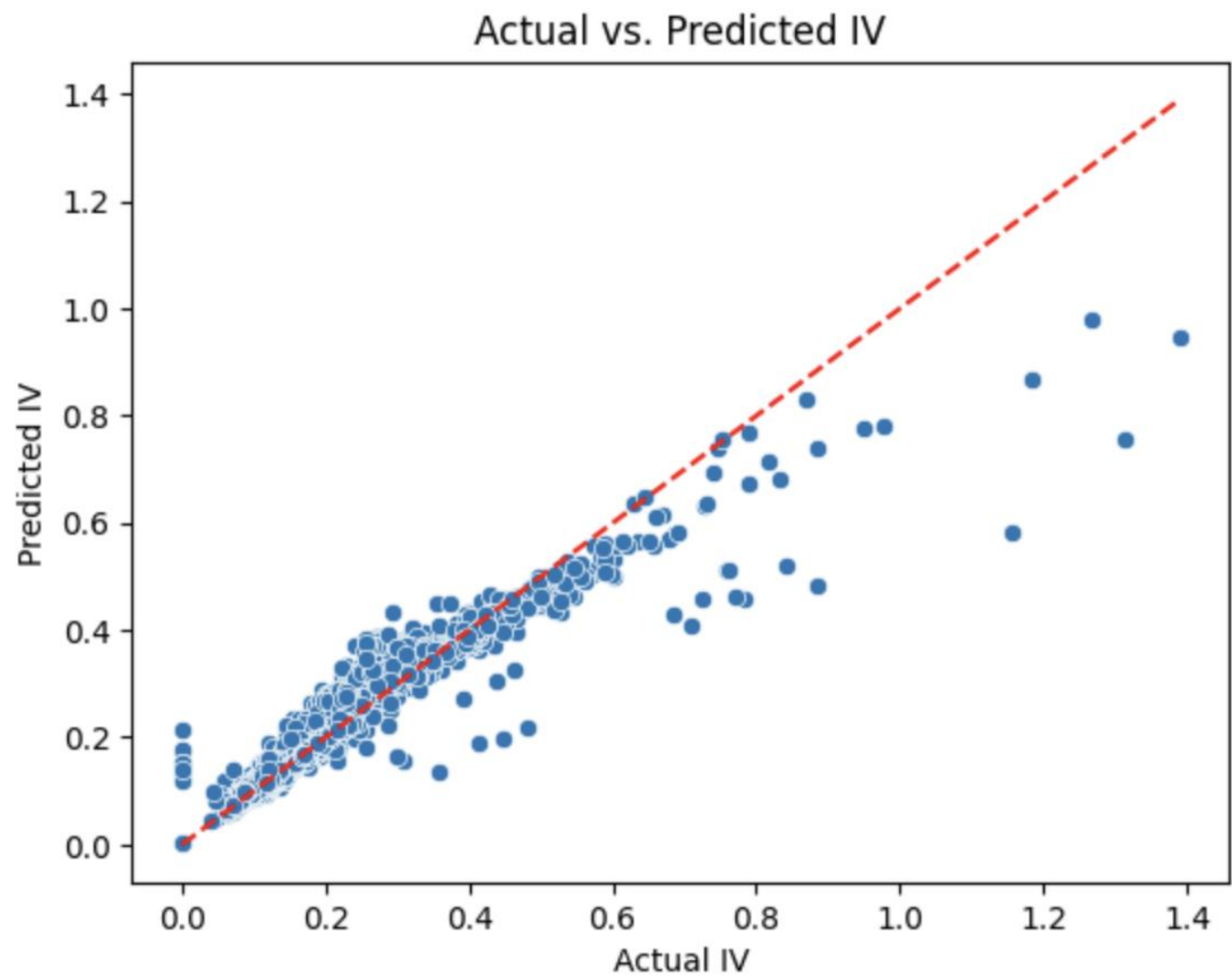
Log-Transformed IV Distribution



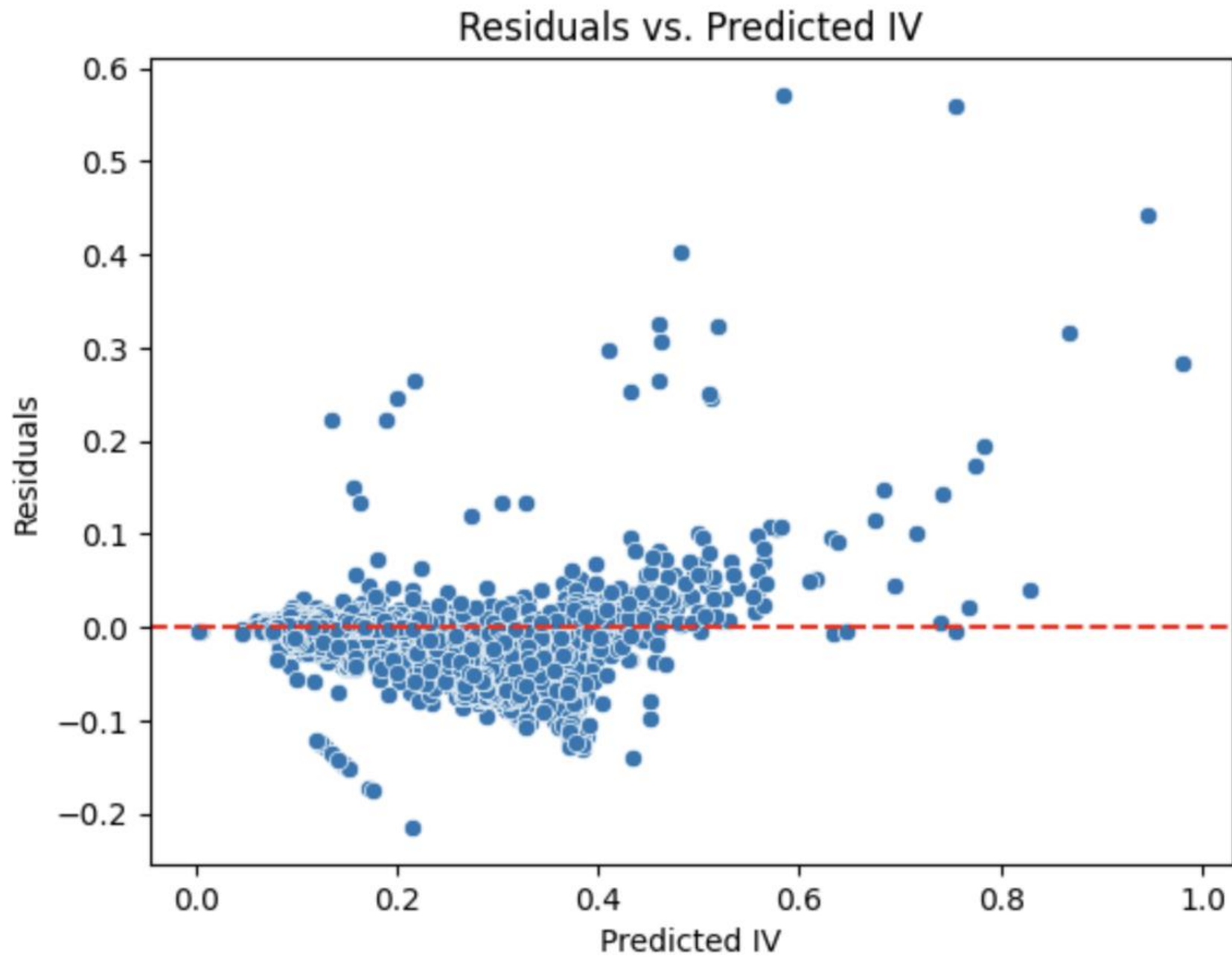
Train using Random Forest Regressor

- MAE: 0.0207917
- MSE: 0.0014346
- R2: 0.86776091
- Important features – strike, c_rho, c_gamma





Scatter plot of Actual vs
Predict log IV



Scatter plot of Residuals
vs Predict log IV

Results and Conclusion

- Low MAE, MSE indicate strong predictive performance
- R^2 below 0.9 indicates inherent uncertainty and noise in data
- Hyperparameter tuning worsened results slightly
- Indicates overfitting, data leakage

Future Work

- Cross Validation while splitting data
- Advanced models using XGBoost, GARCH
- Testing the model on realized volatility
- Backtesting